

Experiment No: 8

Title

Write a Python Program to Implement Depth First Search (DFS)

Aim

To develop a Python program to implement the **Depth First Search (DFS)** algorithm for traversing a graph by visiting each vertex recursively.

Objective

To implement the Depth First Search (DFS) algorithm using recursion to traverse all reachable vertices in a graph from a given starting node.

Algorithm

Step 1: Start the program.

Step 2: Represent the graph using an adjacency list.

Step 3: Create an empty set to store the visited nodes.

Step 4: Select the starting node.

Step 5: Mark the current node as visited and display it.

Step 6: Visit each adjacent unvisited node recursively.

Step 7: Continue the process until all reachable nodes are visited.

Step 8: Stop the program.

Program

```
# Depth First Search (DFS)
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],
```

```
'E': ['F'],  
'F': []  
}
```

```
visited = set()
```

```
def dfs(node):
```

```
    if node not in visited:
```

```
        print(node, end=" ")
```

```
        visited.add(node)
```

```
        for neighbour in graph[node]:
```

```
            dfs(neighbour)
```

```
print("Graph:")
```

```
for node in graph:
```

```
    print(node, "->", graph[node])
```

```
print("\n\nDFS Traversal:")
```

```
dfs('A')
```

Sample Input

Starting Node : A

(The graph is predefined in the program.)

Sample Output

```
Graph:
A -> ['B', 'C']
B -> ['D', 'E']
C -> ['F']
D -> []
E -> ['F']
F -> []
```

```
DFS Traversal:
A B D E F C
```

Result

Thus, the Python program to implement the **Depth First Search (DFS)** algorithm was executed successfully, and all reachable vertices were traversed recursively.

